

# A Comparative Study of Immunomodulatory Organoid Cultures

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## Abstract

We examine the behaviour of tissue phantoms in chronic wound care. The reduced model retains most of the explanatory power at a fraction of the computational cost. Replicate agreement was high, suggesting the protocol is robust to small handling differences. Residual error grows approximately linearly with the number of coupled parameters.

## 1. Introduction

Residual error grows approximately linearly with the number of coupled parameters. Discrepancies at the boundary are consistent with the simplifying assumptions noted above. Sensitivity to the smoothing window proved modest across the range we examined. The measurements reported here were collected over twelve independent trials under controlled conditions. We observe a pronounced inflection once the load exceeds roughly sixty percent of nominal capacity. Longer observation windows narrow the confidence interval but do not shift the central estimate.

Taken together, these findings argue for treating the two regimes separately in future analyses. Neither of the alternative formulations improved on the simple additive model in our setting. We observe a pronounced inflection once the load exceeds roughly sixty percent of nominal capacity. Longer observation windows narrow the confidence interval but do not shift the central estimate. The reduced model retains most of the explanatory power at a fraction of the computational cost. The measurements reported here were collected over twelve independent trials under controlled conditions. Sensitivity to the smoothing window proved modest across the range we examined. A modest but repeatable bias appears in the lowest-magnitude bin and warrants further study.

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work to within the stated tolerance. Neither of the alternative formulations improved on the simple additive model in our setting.

Taken together, these findings argue for treating the two regimes separately in future analyses. We release the processing scripts and intermediate artefacts to support independent verification. The measurements reported here were collected over twelve independent trials under controlled conditions. We observe a pronounced inflection once the load exceeds roughly sixty percent of nominal capacity. Residual error grows approximately linearly with the number of coupled parameters.

## 2. Related Work

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## 3. Materials and Methods

Longer observation windows narrow the confidence interval but do not shift the central estimate. Replicate agreement was high, suggesting the protocol is robust to small handling differences. Sensitivity to the smoothing window proved modest across the range we examined. Taken together, these findings argue for treating the two regimes separately in future analyses. Residual error grows approximately linearly with the number of coupled parameters. The reduced model retains most of the explanatory power at a fraction of the computational cost. Neither of the alternative formulations improved on the simple additive model in our setting. We observe a pronounced inflection once the load exceeds roughly sixty percent of nominal capacity.

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#### **4. Experimental Setup**

Calibration drift over the trial period remained below the threshold set out in the protocol. Residual error grows approximately linearly with the number of coupled parameters. Our baseline reproduces the behaviour described in earlier work to within the stated tolerance. Longer observation windows narrow the confidence interval but do not shift the central estimate. We observe a pronounced inflection once the load exceeds roughly sixty percent of nominal capacity. Taken together, these findings argue for treating the two regimes separately in future analyses.

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#### **5. Results**

Discrepancies at the boundary are consistent with the simplifying assumptions noted above. Calibration drift over the trial period remained below the threshold set out in the protocol. Residual error grows approximately linearly with the number of coupled parameters. We observe a pronounced inflection once the load exceeds roughly sixty percent of nominal capacity. The reduced model retains most of the explanatory power at a fraction of the computational cost. A modest but repeatable bias appears in the lowest-magnitude bin and warrants further study. Neither of the alternative formulations improved on the simple additive model in our setting.

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## 6. Discussion

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